

Ex. 21. A batsman makes a score of 87 runs in the 17th inning and thus increases his average by 3. Find his average after 17th inning.

Sol. Let the average after 17th inning = x .

Then, average after 16th inning = $(x - 3)$.

$$\therefore 16(x - 3) + 87 = 17x \text{ or } x = (87 - 48) = 39.$$

Ex. 22. The batting average for 40 innings of a cricket player is 50 runs. His highest score in an innings exceeds his lowest score by 172 runs. If these two innings are excluded, the average score of the remaining 38 innings is 48 runs. Find his highest score in an innings. (S.S.C., 2006)

Sol. Let the highest score be x . Then, lowest score = $(x - 172)$.

$$\therefore x + (x - 172) = 50 \times 40 - 48 \times 38 \Rightarrow 2x - 172 = 2000 - 1824 = 176$$

$$\Rightarrow 2x = 348 \Rightarrow x = 174.$$

Hence, highest score in an innings = 174.

Ex. 23. 10 years ago, the average age of a family of 4 members was 24 years. Since then, two children having been born, still the average age of the family is the same today. If the two children differ in age by 2 years, find the present age of the younger child. (S.S.C., 2007)

Sol. Total age of 4 members, 10 years ago = (24×4) years = 96 years.

Total age of 4 members now = $(96 + 10 \times 4)$ years = 136 years.

Total age of 6 members now = (24×6) years = 144 years.

Sum of the ages of 2 children = $(144 - 136)$ years = 8 years.

Let the age of the younger child be x years.

Then, age of the elder child = $(x + 2)$ years.

$$\text{So, } x + x + 2 = 8 \Rightarrow 2x = 6 \Rightarrow x = 3.$$

$$\therefore \text{Age of the younger child} = 3 \text{ years.}$$

Ex. 24. Distance between two stations A and B is 778 km. A train covers the journey from A to B at 84 km per hour and returns back to A with a uniform speed of 56 km per hour. Find the average speed of the train during the whole journey.

$$\text{Sol. Required average speed} = \left(\frac{2xy}{x+y} \right) \text{ km/hr} = \frac{2 \times 84 \times 56}{(84 + 56)} \text{ km/hr}$$

$$= \left(\frac{2 \times 84 \times 56}{140} \right) \text{ km/hr} = 67.2 \text{ km/hr.}$$

Ex. 25. The average salary of workers of a factory is ₹ 6000, while the average salary of 150 technicians is ₹ 12000 and that of non-technicians is ₹ 3750. Find the total number of workers in the factory.

Sol. Let the total number of workers be x . Then,

$$6000x = (12000 \times 150) + 3750(x - 150) \Rightarrow 2250x = 1237500 \Rightarrow x = 550.$$

Hence, total number of workers in the factory = 550.

Ex. 26. The average score of girls in class X examination in a school is 73 and that of boys is 71. The average score in class X of that school is 71.8. Find the percentage of the number of girls and boys in class X of the school. (Campus Recruitment, 2006, 2009; M.A.T., 2003, 06; S.S.C., 2008)

Sol. Let the number of boys in the class be x and the number of girls be y .

$$\text{Then, } 71x + 73y = 71.8(x + y) \Rightarrow 0.8x = 1.2y \Rightarrow x = \frac{3}{2}y.$$

$$\text{Percentage of boys} = \left[\frac{x}{(x+y)} \times 100 \right] \% = \left[\frac{\frac{3}{2}y}{\left(\frac{3}{2}y + y\right)} \times 100 \right] \%$$

$$= \left(\frac{3y}{2} \times \frac{2}{5y} \times 100 \right) \% = 60\%.$$

$$\text{Percentage of girls} = (100 - 60)\% = 40\%.$$